INCH-POUND

MIL-C-11418H
30 June 1992
SUPERSEDING
MIL-C-11418G
14 August 1980

MILITARY SPECIFICATION

CLAMPS, LEAK REPAIR, PIPE COUPLING:

FOR USE OVER GROOVED-TYPE COUPLINGS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the repair clamps to be used to provide leakproof, flexible housing over leaking couplings of grooved-end pipe.

1.2 Classification. Clamps shall be the following nominal pipe sizes, as specified (see 6.2):

4-inch
6-inch
8-inch
12-inch

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730

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2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-10387	- Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gasket for Grooved-End Pipe and Tube.

STANDARDS

FEDERAL

FED-STD-595	- Colors for Use in Government Procurements.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-210	- Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Phailadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 47	- Ferritic Malleable from Castings.
A 183	- Carbon Steel Track Bolts and Nuts.
A 193	- Steel, Stainless and Alloy Steel Bolting Materials for High Temperature Service.
A 194	- Nut for Bolts, Carbon and Alloy Steel for High Pressure and High Temperature Service.
A 536	- Castings, Ductile Iron.
B 26	- Aluminum-Alloy Sand Castings.
B 633	- Zinc on Iron or Steel, Electrodeposited Coatings of.
D 395	- Rubber Property - Compression Set.
D 412	- Rubber Properties in Tension.
D 471	- Rubber Property - Effects in Liquids.
D 573	- Rubber Deterioration in an Air Oven.
D 1053	- Measuring Rubber Property - Stiffening at Low Temperatures Using a Torsional Wire Apparatus.
D 1229	- Rubber Property - Compression Set at Low Temperature.
D 2240	- Rubber Property - Durometer Hardness.
D 3951	- Standard Practice for Commercial Packaging.
D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The repair clamps shall be designed to clamp over the couplings specified in MIL-C-10387 and form a tight seal on the pipe surfaces on both sides of the coupling, thereby containing any liquid leaking from the pipe-coupling joint area. The repair clamp also shall fit over any portion along the length of pipe to prevent leakage over a small area such as a bullet hole.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The repair clamp shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments (-25 to +160 °F) to which the repair clamp may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Design. The repair clamp shall be designed to fit over the coupling-conforming to MIL-C-10387. The interior of the clamp shall provide a clearance of not less than 1/4-inch over all parts of the coupling. If a vent hole is drilled and tapped in to the top portion of the clamp housing, then it must be fitted with a threaded pipe plug. Seals shall be factory installed in their proper position in the repair clamp. Bolts shall be designed to permit tightening of nuts using a 10-inch or 12-inch adjustable wrench without undue interference.

3.5 Clamp housing. The repair clamp housing shall be cast aluminum alloy UNS A 03560 in accordance with ASTM B 26 in the T6 temper, cast ductile iron conforms to ASTM A 536 (grade 65-45-12) or ASTM A 47 (grade 35018) or any other material subject to Government approval. The repair clamp shall be reusable, excluding gasket.

3.6 Pressure ratings. The repair clamps shall seal the applicable pressure specified in table I. The clamp shall not leak when tested with water at the hydrostatic pressures specified in table I.

TABLE I. Service pressure ratings.

Size	Working Pressure	Hydrostatic Test Pressure
(inches)	(psi)	(psi)
4	600	1200
6	600	1000
8	450	900
12	300	600

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3.7 Seals/gaskets. Gaskets shall be suitable for use with diesel fuels, jet fuels or automotive gasoline. Seals or gaskets shall be molded from a synthetic rubber compound. The seals and gaskets shall conform to the requirements specified in table II.

TABLE II. Physical requirements of the seals/gaskets and corresponding test methods.

Description	Requirement	ASTM Test Method
Original properties		
Tensile strength, PSI	1,500 min.	D 412
Ultimate elongation, %	300 min.	D 412
Hardness, Shore A Durometer, points	60 ±0	D 2240
After oven aging for 22 hours at 158 °F		
Tensile strength retained, % min.	65	D 573
Elongation retained, % min.	65	D 573
Hardness changed, Shore A, points max.	±15	D 2240
Compression set, % max.	50	D 395
After immersion in ASTM Reference Fuel B for 94 hours at 73.4 °F		
Tensile strength retained, % min.	40	D 471
Elongation retained, % min.	60	D 471
Volume change %	-5 to +35	D 471
After immersion in ASTM Reference Fuel A for 94 hours at 73.4° F volume change, %	-5 to +15	D 471
Low temperature properties after extraction in Reference Fuel B per ASTM D 471 for 94 hours at 73 °F, dried in air circulating oven at 130 °F for 72 hours and cold conditioning for 94 hours at -25 °F.		
Compression set at 40 % deflection, % max.	50	D 1229
Torsional stiffness ratio, max.	10	D 1053

3.8 Bolts and nuts. Bolts shall conform to ASTM A 183, grade 2 or shall be in accordance with ASTM A 193, grade B7. Nuts shall be heavy hexagon type and in conformance with ASTM A 194, grade 2H. Bolts and nuts shall be zinc plated in accordance with ASTM B 633, type II, classification Fe Zn 13.

3.9 Treatment and painting. All exterior surfaces of the clamps except the gaskets, bolts and nuts shall be cleaned, treated and painted in accordance with MIL-T-704, type as applicable. Unless otherwise specified (see 6.2),

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high zinc primer and superior quality enamel top coat, used by the manufacturer of the commercial item, shall be used. The top coat color shall be in accordance with color no. 33446 of FED-STD-595.

3.10 Identification marking. The clamps shall be permanently identified in accordance with MIL-STD-130. Each body half shall include the appropriate working pressure rating as specified in table I as follows: "Working pressure to _____ psi".

3.11 Workmanship. All parts of the clamps shall be free of burrs, sharp edges, or other defects that could impair operation or serviceability, or be injurious to operating and maintenance personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).,
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 First article examination. The first article repair clamps shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article repair clamps shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, special inspection level S-1, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.1.1 Lot. A lot shall consist of all clamps of the same size offered for inspection at one time, not to exceed 150 repair clamps.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. One or more defects shall be cause for rejection of the lot.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The repair clamps shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
105. Design not as specified (see 3.4).
106. Clamp housing not as specified (see 3.5).
107. Pressure rating not as specified (see 3.6).
108. Seals/gaskets not as specified (see 3.7).
109. Bolts and nuts not as specified (see 3.8).
110. Treatment and painting not as specified (see 3.9).
111. Identification marking not as specified (see 3.10).
112. Workmanship not as specified (see 3.11).

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4.5.2 Tests.

4.5.2.1 Seals/gaskets. The seals/gaskets shall be tested in accordance with the ASTM test methods specified in table II. Nonconformance to the requirements of table II shall constitute failure of this test.

4.5.2.2 Hydrostatic. The repair clamps shall be subjected to the hydrostatic pressures specified in table I. The pressure shall be held for not less than 10 minutes, using water as the test fluid. Examine clamps for leakage, rupture, or yielding. Evidence of any leakage, rupture, or yielding shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when one or more defects are found.

4.6.1.3 Examination.

4.6.1.4 Samples. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

113.	Preservation not as specified for level A (see 5.1.1).
114.	Packing boxes not as specified for level A or level B packing (see 5.2.1 and 5.2.2).
115.	Strapping not as specified for level A or B (see 5.2.1 and 5.2.2).
116.	Preservation and packing not in accordance with the reference documents for commercial (see 5.1.2 and 5.2.3).
117.	Marking and packing lists not as specified (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each complete repair clamp with instruction card shall be placed in a snug fitting polyethylene bag which shall be closed by heat sealing. Bag material shall conform to L-P-378, type II, minimum of 6 mils thick. Prior to placement of the clamp in the bag, surfaces of the seals/gaskets that can contact each other shall be protected by placing wax paper or a narrow strip of plywood to prevent deformation of seals/gaskets and protection against sticking. The follower bolts of the assembled clamp shall be drawn fingertight. Unprotected surfaces of the body, bolts, and nuts shall

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be coated with type P-1 preservative conforming to the applicable specification listed in and applied in accordance with MIL-P-116.

5.1.1.1 Instruction card. Each instruction card shall be attached to the clamp in a protected location.

5.1.2 Commmercial. Each complete clamp with instruction card shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The clamps preserved as specified in 5.1, shall be packed in a close fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional. Blocking and bracing shall be provided to prevent free movement and damage to the gasket. Only clamps of same size and like description shall be packed together. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable per ASTM D 4675.

5.2.2 Level B. The clamps preserved as specified in 5.1, shall be packed as specified in 5.2.1 except that the boxes shall be domestic type.

5.2.3 Commercial. The clamps shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for the military levels of protection (level A or level B) shall be in accordance with MIL-STD-129. Both interior and exterior packing lists shall be provided.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The repair clamp is intended for use over couplings conforming to MIL-C-10387 in repairing leaks during operation of a pipeline.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Size of pipe (see 1.2).
c. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents reference (see 2.1.1 and 2.2).
d. When a first article is required for inspection (see 3.2).

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e. Primer and top coat color other than as specified (see 3.9).
f. Levels of preservation and packing required (see 5.1.and 5.2).
g. Any special markings (see 5.3).

6.3 First article. When first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Flexible housing
Over coupling
Pipe
Pipeline
Seals/gaskets

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activities:
Navy - MC
DLA - CS

User activity:
Army - CE

Preparing activity:
Army - ME

Project 4730-0034

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-11418H	2. DOCUMENT DATE (YYMMDD) 920630

3. DOCUMENT TITLE
Clamps, Leak Repair, Pipe Coupling: For Use Over Grooved-Type Couplings

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code) Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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